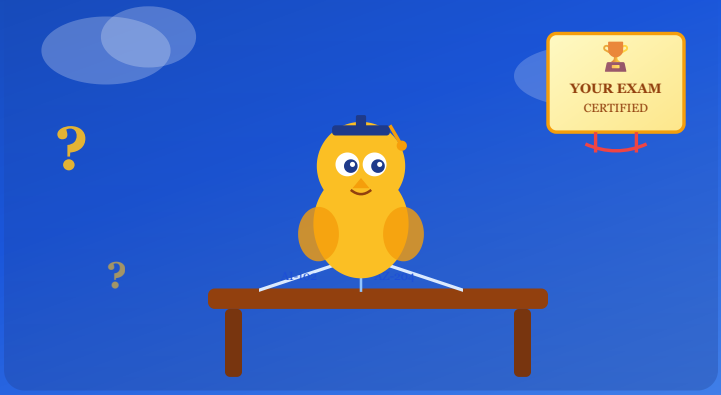
THE ILLUSTRATED

Children's Guide to Cert Buddy

A tale of study, practice, and passing exams



Written for every learner who ever stared at a certification outline and wondered, "Now what?"

Cert Buddy

Cert Buddy

Brought to you by the Cert Buddy generator template

Inspired by *The Illustrated Children's Guide to Kubernetes* by Matt Butcher (DEIS).
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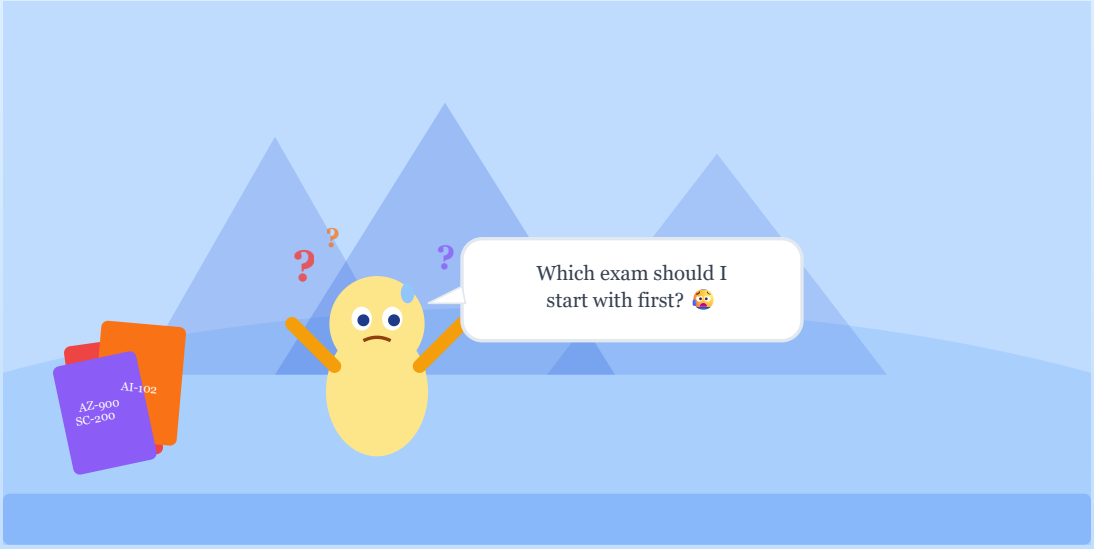
Character "Certy the Owl" is original to this guide.

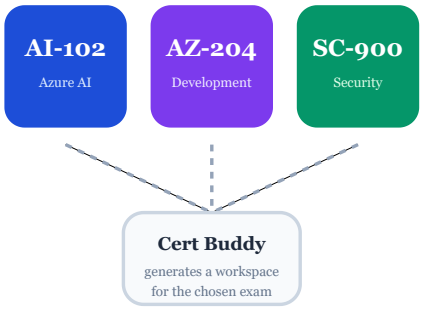
Dedicated to every engineer who has opened Microsoft Learn at midnight
the week before their certification exam.



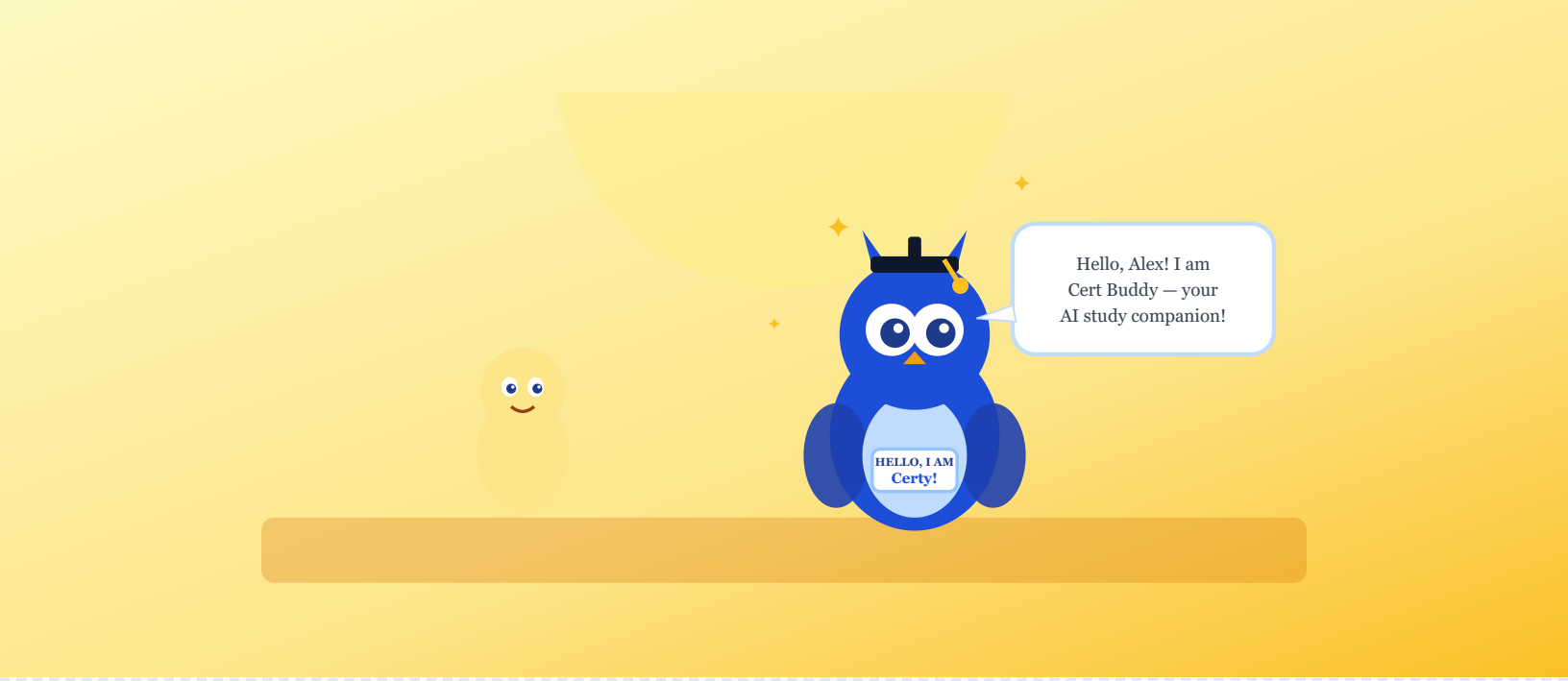
Once upon a time, there was a learner named Alex. Alex wanted to earn **a Microsoft certification**, but could not decide whether to start with AI-102, AZ-204, SC-900, or something else entirely. Every exam had a different skills outline, different product names, and a different style of practice. Alex stared at the screen, surrounded by a tower of study guides, and whispered, "Where do I even begin?"

Pick an Exam

- **Works for any Microsoft certification exam**
  Azure, data, security, Microsoft 365, fundamentals, and more
- **Starts with the exam code and official name**
  Examples: AI-102, AZ-204, SC-900
- **Builds a dedicated workspace for that exam**
  Agent, prompts, skills, references, and MCP configuration
- **Adapts the content to the exam family**
  Hands-on Azure labs for Azure exams; more conceptual focus for fundamentals
- **AI-102 can still be used as a concrete example**
  This guide uses AI-102 examples after the generic setup



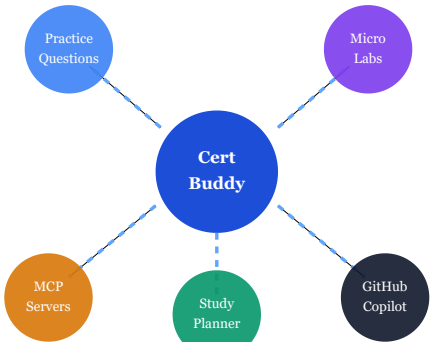
Cert Buddy does not lock you into a single exam. You choose the certification code, the official exam name, and the domain, and Cert Buddy generates a workspace tailored to that exam. In the rest of this guide, AI-102 is used as the running example, but the same structure works for AZ-204, SC-900, DP-203, and other Microsoft certification paths.



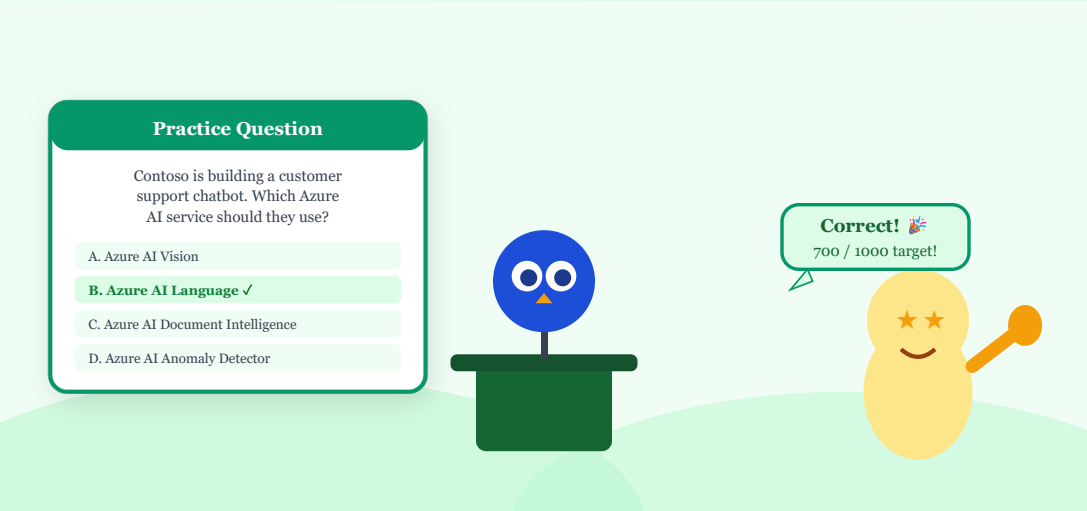
Just as Alex was about to close the laptop and give up, a friendly owl appeared on the screen. "Hello there, learner!" the owl said cheerfully, adjusting the tiny graduation cap on its head. "My name is **Certy**, and I am your Cert Buddy — a study companion powered by GitHub Copilot. I can help you prepare for any Microsoft certification exam by generating an exam-specific workspace, then guiding you with practice questions, hands-on labs when the exam supports them, and a personalised study plan. Shall we begin?"

Cert Buddy

- **Powered by GitHub Copilot + MCP servers**
- **Three core capabilities:**
  Practice Questions · Micro-Labs · Study Planner
- **Grounded in Microsoft Learn documentation**
- **Works for any Microsoft certification**
- **Lives right inside VS Code**



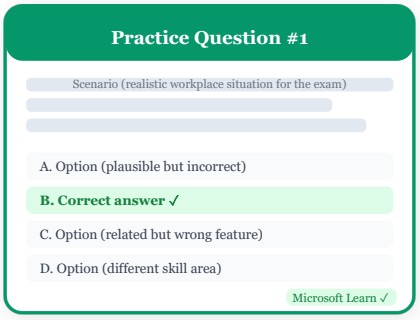
Cert Buddy is an AI-powered study companion that lives inside VS Code. It uses GitHub Copilot, exam-specific prompts and skills, and MCP servers to ground its output in current documentation. First it generates a workspace for the exam you choose. Then that workspace gives you one exam-specific buddy agent with three specialist helpers for questions, labs, and study planning.



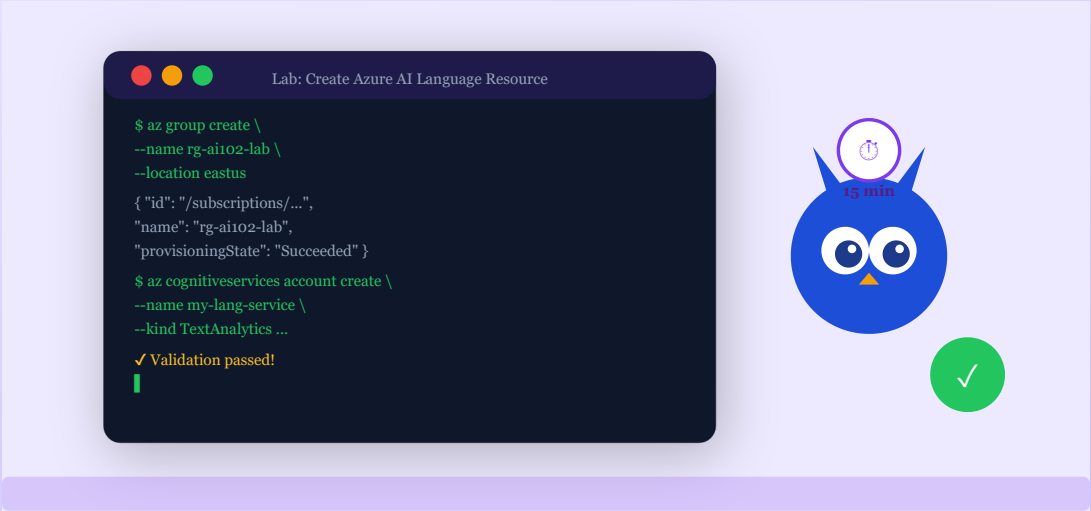
"First," said Certy, "let us practise some questions." For this guide, Certy chose **AI-102** as the example exam. A card appeared, describing a scenario at a fictional company called Contoso. "Read the scenario carefully," Certy advised, "then choose the best answer — there is always exactly one correct option." Alex answered and heard, "Correct! Azure AI Language powers question answering and conversational language understanding — ideal for chatbots." With each question, Alex began to see patterns.

Practice Questions

- **Scenario-first format**
  Every question opens with a workplace scenario
- **Exactly 4 options (A–D)**
  Plausible distractors use real services or features for the chosen exam
- **One correct answer per question**
  Tests knowledge, not trick wording
- **Detailed rationale after you answer**
  Explains why each option is right or wrong
- **Grounded in Microsoft Learn**
  Via Context7 MCP — with current terminology for that certification



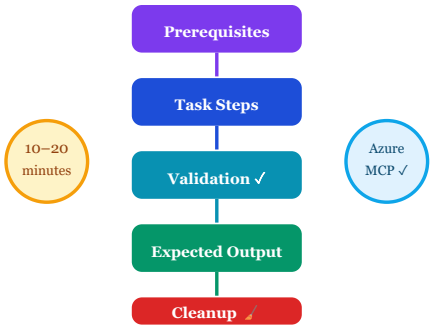
Every practice question follows strict exam-style rules: a workplace scenario comes first, then four plausible options using real product names from the chosen exam domain. After you respond, Cert Buddy reveals the correct answer with a full rationale — explaining not just which option is right, but *why* the others are wrong. Questions are generated on demand, so you never see the same question twice.



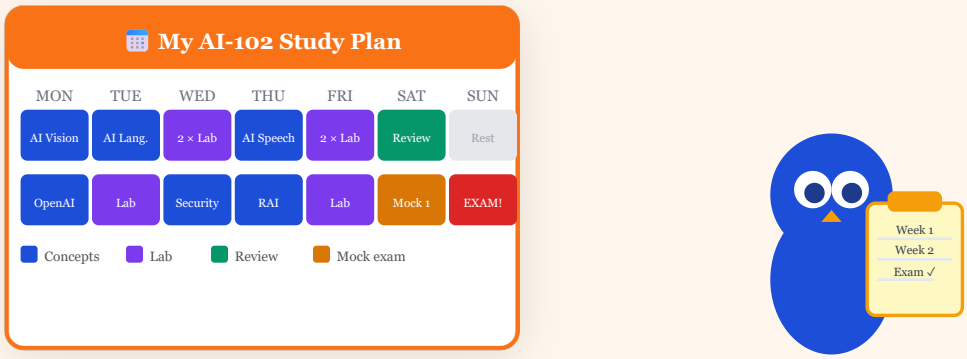
"Questions are great," said Alex, "but I learn by doing." Certy nodded wisely. "Then let us run a **hands-on lab**." For exams like AI-102 and AZ-204, Cert Buddy can generate short, validated labs. In this example, the terminal window shows an AI-102 lab that creates a real Azure AI Language resource. Each step has an explicit validation check, so Alex knows exactly when the task is done. "And do not worry," Certy added, "there are cleanup steps at the end so you do not accidentally leave resources running up charges."

Hands-on Labs

- **10–20 minutes each**
  Single, focused learning objective per lab
- **Available when the exam supports hands-on work**
  Especially useful for Azure exams such as AI-102 and AZ-204
- **Real commands and validation**
  Azure exams use Azure MCP for accuracy checks
- **Explicit validation steps**
  Know exactly when each task succeeds
- **Cleanup instructions included**
  No surprise Azure charges after the lab
- **Cost warnings upfront**
  Free tier where possible; charges flagged clearly



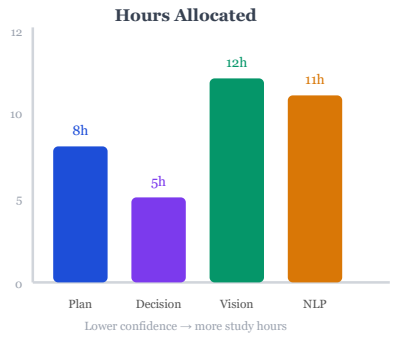
A hands-on lab is a short exercise with a single learning objective. In generated workspaces for practical exams, each lab follows the same structure: prerequisites, numbered task steps, explicit validation checks, expected command output, and cleanup. For Azure exams, commands are validated with Azure MCP before delivery, so you can trust they work on the current version of the Azure CLI.



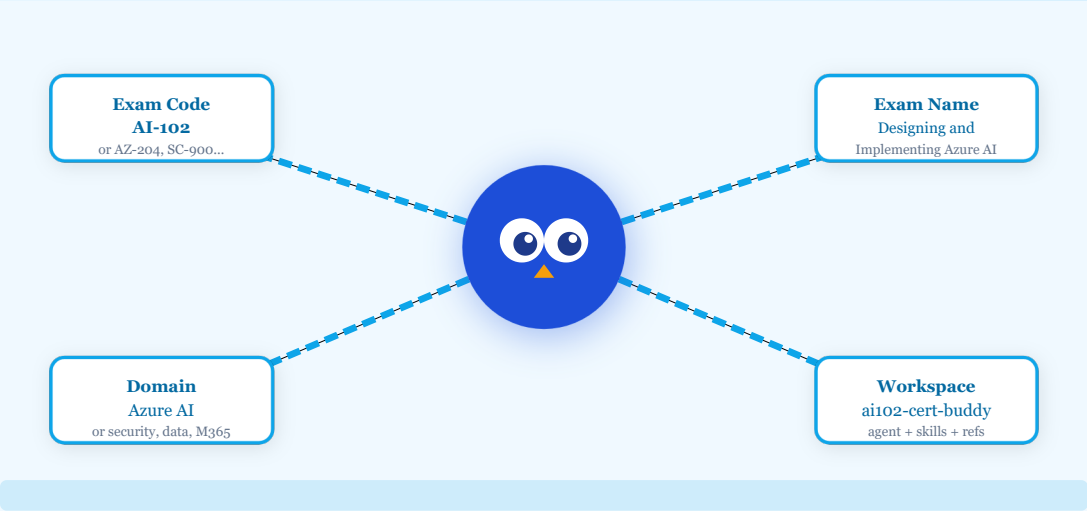
After answering a set of confidence questions, Certy appeared with a clipboard. "Based on your answers, I have built you a **personalised study plan**," Certy announced. "Whatever exam you choose, I can rank the weak areas first, estimate the hours, and link each topic to the right Microsoft Learn material." In this guide, the sample plan is for AI-102, so the calendar includes Azure AI Vision, Azure AI Language, and Azure OpenAI Service topics. Alex looked at the two-week calendar and nodded. It felt achievable.

The Study Planner

- **Confidence assessment first**
  Rate yourself on each exam skill area (1–5)
- **Weak areas prioritised**
  Low-confidence, high-weight topics get more time
- **Estimated study hours**
  Per skill area, based on weight and your confidence
- **Microsoft Learn links**
  Direct module links for every topic
- **Day-by-day schedule**
  Balanced across your available time before exam



The study planner begins with a short self-assessment: you rate your confidence across the exam skill areas. Cert Buddy then builds a schedule that prioritises low-confidence, high-weight topics, estimates the hours needed, and links each session directly to the matching Microsoft Learn module — so you always know exactly what to study next.



"How do I make this work for a specific exam?" Alex asked. Certy twirled, and four glowing boxes lit up: **exam code**, **exam name**, **domain**, and **workspace**. "First, you ask the generator to build an exam-specific Cert Buddy," Certy explained. "For example, you can create an AI-102 workspace, an AZ-204 workspace, or an SC-900 workspace. The structure stays the same, but the content changes to match the exam."

Generated Structure

- **Start with the generator agent**
  `@cert-buddy-generator Create a study buddy for AI-102`
- **Or use the Python script**
  `python generate_cert_buddy.py --exam AI-102 --name "Designing and Implementing a Microsoft Azure AI Solution" --domain "Azure AI" --output ../ai102-cert-buddy`
- **The generated workspace always includes one exam agent**
  `@ai102-cert-buddy-agent` for the AI-102 example
- **And three specialist helpers inside that workspace**
  Item creator, lab creator, and study planner
- **Folder structure stays consistent across exams**
  Agent, prompts, skills, references, and MCP config

The generator creates the same high-quality shape for every exam: one exam-specific buddy agent, two prompt templates, three specialist helpers, reference files, and MCP configuration. The names change to match the exam code, so AI-102 becomes `ai102-cert-buddy-agent`, `ai102-item-creator`, `ai102-lab-creator`, and `ai102-study-planner`.

Once the AI-102 workspace was generated, Certy showed Alex the three specialist helpers inside it: **item creator** for practice questions, **lab creator** for short hands-on exercises, and **study planner** for a personalised schedule. "You speak to one exam-specific buddy agent," said Certy, "and it routes the work to the right helper for you." Alex grinned. The setup finally made sense.

Examples

What to do	Example
Generate an exam workspace	`@cert-buddy-generator Create a study buddy for AI-102`
Generate another exam workspace	`@cert-buddy-generator Create a study buddy for AZ-204`
Generate with Python instead	`python generate_cert_buddy.py --exam AI-102 --name "Designing and Implementing a Microsoft Azure AI Solution" --domain "Azure AI" --output ../ai102-cert-buddy`
AI-102 item creator example	`@ai102-cert-buddy-agent give me 5 practice questions on Azure AI Vision`
AI-102 lab creator example	`@ai102-cert-buddy-agent create a 15-minute lab on conversational language understanding`
AI-102 study planner example	`@ai102-cert-buddy-agent help me create a study plan for AI-102`

How the three helpers map to the AI-102 example: `ai102-item-creator` powers the practice question request, `ai102-lab-creator` powers the lab request, and `ai102-study-planner` powers the study plan request. You talk to `@ai102-cert-buddy-agent`, and it chooses the right helper automatically.